MSCA 560: Distributed Systems and Network Security

Individual Term Project — Fall 2026

Purpose

Choose a focused security problem involving a network or distributed system. Your project should lead to a question you can answer through implementation, measurement, or testing. You may build something, investigate an existing system, or reproduce a published result.

A successful project does more than describe a technology or demonstrate that a tool runs. It explains what security problem matters, how you investigated it, what you found, and how confident we should be in your conclusion. A well-explained result that differs from your expectation is valuable.

The project is individual and accounts for 20% of the course grade.

Choose a project path

You may propose your own topic under any of these paths:

1. **Analyze.** Assess the security of a defined network or distributed-system design. Identify assets, trust boundaries, plausible threats, and the effectiveness of existing controls. Support your assessment with evidence such as configuration review, a policy test, packet analysis, or a documented case comparison.
2. **Measure.** Investigate a security question through repeatable observations. For example, compare how two configurations affect detection, connection behavior, or service availability in a controlled environment. State your measures and explain sources of uncertainty.
3. **Innovate.** Design a security mechanism, policy, or architecture and evaluate whether it addresses a specific threat. You may implement a small prototype. Keep the claim proportionate to what you can test during one semester.
4. **Test or reproduce.** Reproduce a focused result from a published paper, technical report, or documented security experiment. Record what you could reproduce, what differed, and possible reasons for the difference. You are not expected to replicate an entire large study.

Possible topics include secure communication between services, authentication and access control, intrusion detection, network segmentation, TLS or VPN configuration, secure healthcare data exchange, cloud or container security, and defenses against attacks studied in class. Your final project must go beyond repeating a course lab.

For ideas, you are encouraged to browse the last several years proceedings of the 4 top-tier security conferences:

- USENIX Security

- ACM CCS
- IEEE Security and Privacy
- NDSS

Note: You might also find this blog post by Zhiyun Qian very helpful.

Project stages

Stage	What to submit	Due date
Proposal	A 1–2 page proposal, excluding references	October 4, 2026
Progress check	A brief update with work completed, initial evidence, obstacles, any justified change in scope, and a link to your GitHub repository	October 25, 2026
Presentation and feedback	A 10–15 minute presentation followed by questions	November 9, 2026
Final submission	Final report, GitHub repository link, and supporting materials	November 12, 2026

The presentation takes place before the final submission so that you can use questions and feedback to strengthen your report.

Proposal requirements

Submit a **1–2 page proposal, excluding references**, by **October 4, 2026**. Choose the list that best fits your project.

If you will investigate or test a security issue, include:

- **Title and research question:** What do you want to find out?
- **System and scope:** What will you study?
- **Method:** What will you analyze, measure, or reproduce?
- **Evaluation:** What evidence will you collect, and how will you interpret it?

If you will build a security solution, include:

- **Title and problem:** What security weakness will you address?
- **Solution idea:** What will you build or configure?
- **Approach:** What tools and steps will you use?
- **Evaluation:** How will you test whether it works?

For either type, briefly explain why the topic matters, list initial references, and give a realistic plan through the final submission.

GitHub repository

Create a GitHub repository for your individual project and update it as your work progresses. Provide the repository link with your progress check and final submission. The repository must be accessible to the instructor for grading; if it is private, grant the instructor access before submitting the link.

Include the materials relevant to your project, organized so that someone else can follow your work:

- A `README.md` stating your project question, approach, repository contents, and instructions for examining or reproducing your results.
- Your original code, scripts, notebooks, configuration files, and test cases, **if your project uses them**.
- Results such as tables, figures, permitted packet captures, or output from your experiments, with enough context to explain how they were produced.

Commit work as you make progress rather than uploading everything at the end. You may use a private repository. **Do not upload credentials, patient information, sensitive data, or material you do not have permission to share.** Cite sources and link to published papers rather than uploading copyrighted copies of them.

Final report

Write a technical paper of **6–8 pages, including references**, using an IEEE conference paper template. Your report should contain an abstract, an introduction with the problem and your contribution, relevant prior work, your method or design, results, a discussion of limitations, a conclusion, and references. Use section headings that fit your particular project.

Submit the report as a PDF in Blackboard and provide the link to your GitHub repository. The report should point readers to the relevant files in the repository when discussing your methods or results. Cite software, datasets, figures, and ideas that are not your own.

Presentation

In **10–15 minutes**, explain the problem, your approach, your most important evidence, and what you concluded. Allow time afterward for questions. If you wrote code or configured a system, be prepared to explain your design decisions and demonstrate the relevant behavior.

After your presentation, note at least one question or piece of feedback that you considered while preparing the final report.

Responsible project work

Perform security tests only on systems and data you own or are explicitly authorized to use. Use the isolated course lab or another approved environment for attack and defense experiments. Do

not test campus, public, clinical, or third-party systems without written authorization and my approval of the project scope. Do not include credentials, patient information, or other sensitive data in your report or GitHub repository.

Your analysis, writing, and interpretation must be your own. If you use a generative AI tool for brainstorming, coding help, or editing, include a brief disclosure describing its role. Check its output, verify every reference, and remain prepared to explain all submitted work.

Resources

- IEEE conference paper templates